Adarsh Varshney

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LinkedIn, GitHub

Software engineer with two years in NVIDIA's Power Architecture Analysis team, building the internal tools its engineers run daily. Rebuilt the team's power-analysis platform in Rust and React (175x faster runs, adopted by 40 engineers) and built PACE, an award-winning RAG chat engine over the team's domain knowledge. Ships full-stack, works AI-first with Claude Code, and deploys edge AI systems as a freelancer. B.Tech in Computer Science from NIT Andhra Pradesh with a Springer-published project on reinforcement learning.

Core Competencies

- **Languages & Frameworks**: Python, Rust, TypeScript/React, SQL, FastAPI; earlier work in Java, Kotlin, C and C++.
- **GenAI & RAG**: Retrieval pipelines and agentic workflows with LiteLLM, ChromaDB, OpenWebUI; daily AI-assisted engineering with Claude Code and peer tools (Codex CLI, Cursor).
- **Edge AI**: Vision-model optimization and deployment with ONNX and OpenVINO on Intel NUC hardware.
- **Platforms & Tooling**: Linux, Windows/WSL2, Docker, AWS, MySQL, MongoDB, Git.
- **Domain**: Silicon power architecture analysis; internal developer tooling with measured adoption.

Professional Experience

NVIDIA — **Bengaluru, India**
ASIC Engineer — *2024–Present*

- Rebuilt the team's power-analysis web platform with a Rust backend and React frontend, cutting per-run analysis time from 1m45s to 600ms (about 175x) and reducing modelling effort 60x; adopted by a team of 40 engineers.
- Built PACE (Power Architecture Chat Engine), a RAG chatbot over fragmented codebase and Confluence knowledge (OpenWebUI, ChromaDB, LiteLLM, FastAPI); used by 12 interns during ramp-up with 9.5/10 feedback and used to surface and refresh stale documentation.
- Designed PACE as a foundation for agentic workflows over the team's domain knowledge; won NTech 2025, NVIDIA's internal paper presentation competition.
- Recognized as a top contributor on the team in the FY2025 performance cycle.
- Work AI-first: daily agentic development with Claude Code across the internal-tooling stack.

Independent — **Remote / Hyderabad, India**
Freelance Edge AI Developer — *2024–Present*

- Built an image-based age/gender classification model on public datasets and deployed it on Intel NUCs (ONNX, OpenVINO) for retail visitor analytics and targeted advertising.

MRM Procom Pvt. Ltd. — **India**
Android Developer — *2022*

- Designed and developed Procom Smart Devices, an Android app to connect, control, and manage Procom devices: real-time monitoring, parameter modification, data logging and export (Java).
- Achieved 100% test coverage with Espresso against the SRS; awarded a Certificate of Recommendation.

Testright Nanosystems Pvt. Ltd. — **India**
Software Developer Intern — *2022*

- Cut R&D time by 3–4 weeks with a custom Camera2API camera module; improved diamond-classification accuracy by 23% via a parameterized pixel-by-pixel comparison algorithm (OpenCV, Java).

- **NIT Andhra Pradesh** — **Tadepalligudem, India**
  Teaching Assistant — *2021–2024*
  - Ran Data Structures & Algorithms and Competitive Programming labs for 120 students; delivered bi-weekly lectures rated 4.8/5.

Education

- **National Institute of Technology, Andhra Pradesh** — **Tadepalligudem, India**
  B.Tech in Computer Science Engineering, Minor in AI for Signal Processing — *2020–2024*
  In-major CGPA 8.66/10. Final-year project on knowledge-infused reinforcement learning for StarCraft II, published as a Springer chapter (see Publications).

Languages

- Hindi (native), English (fluent), Telugu (spoken comprehension).

Publications

- Varshney, A., Seshadri, K., Andraju, V.K. (2026). Assessment of the Knowledge-Infused Policy Gradient Approach for StarCraft. *ICT for Intelligent Systems*, Springer. DOI link

Honors and Awards

- **Winner, NTech 2025** – NVIDIA internal paper presentation competition, for PACE.
- **Economic Times Campus Star** – 87 selected from 35,000 participants across 2,000+ engineering colleges (2023).
- **IEEEXtreme 16.0** – Country Rank 31 of 14,683 participants (2022).

References

- Available upon request.